

Principal Component Analysis



Prithwijit Guha
Dept. of EEE, IIT Guwahati

Feature Selection

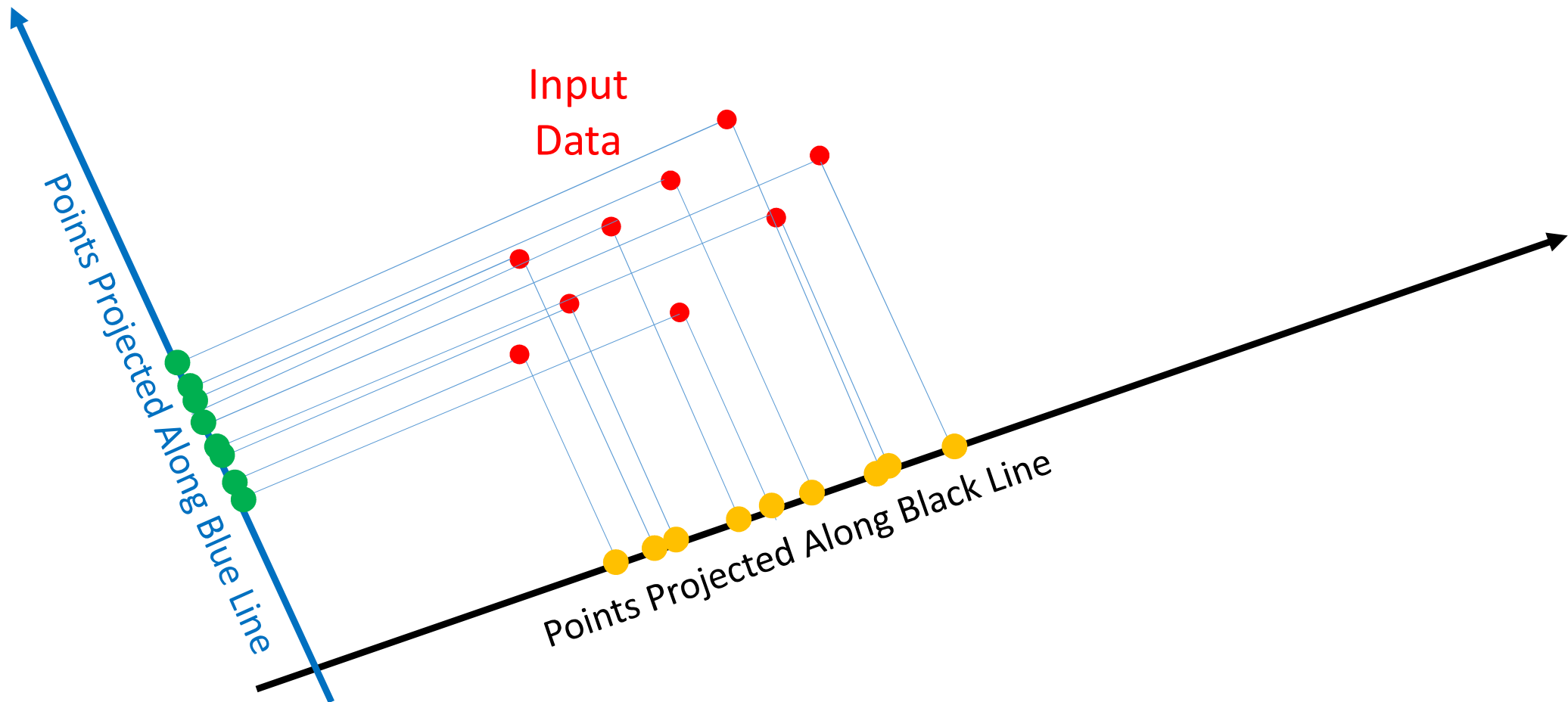
6	-4	4	7	0	-7	1	-8	0	-7
8	-9	-4	-5	0	8	10	-1	6	7
6	5	7	7	6	5	4	8	5	6
1	0	1	0	1	0	1	0	1	0

$$d = 1: \sigma_1 = 5.51$$

$$d = 2: \sigma_2 = 6.63$$

$$d = 3: \sigma_2 = 1.14$$

Principal Components



The Projected Scalars

$$\mathbf{X} = \{\mathbf{x}_i : \mathbf{x}_i \in \mathbb{R}^D; i = 1, \dots, n\}$$

$$\mathbf{Z} = \{z_i : z_i = \mathbf{p}^T \mathbf{x}_i; \forall \mathbf{x}_i \in \mathbf{X}; z_i \in \mathbb{R}^1\}$$

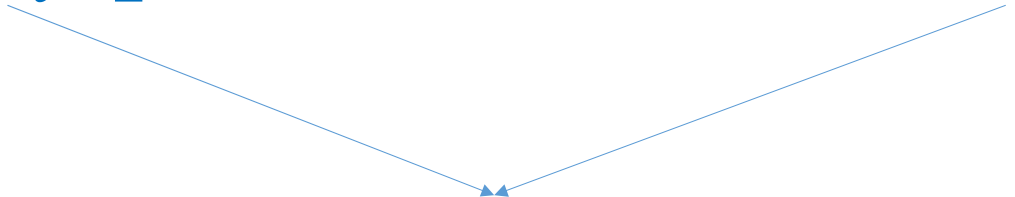
$$\mathbf{p}^T \mathbf{p} = 1$$

The Direction $\mathbf{p}$ is to be optimally chosen such that the variance of $\{z_i; i = 1, \dots, n\}$ is maximized

Evaluating Mean of $\mathbf{Z}$

$$\gamma = \frac{1}{n} \sum_{i=1}^n z_i$$

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$


$$\gamma = f(p, \mu)$$

Evaluating Mean of $\mathbf{Z}$

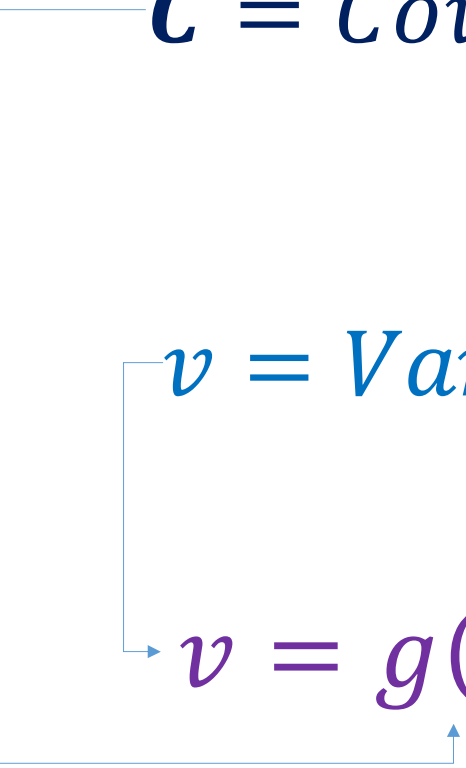
$$\gamma = \frac{1}{n} \sum_{i=1}^n z_i = \frac{1}{n} \sum_{i=1}^n \mathbf{p}^T \mathbf{x}_i = \mathbf{p}^T \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \right)$$

$$\gamma = \mathbf{p}^T \boldsymbol{\mu}$$

Evaluating Variance of $\mathbf{Z}$

$$\mathbf{C} = \text{Cov}(\mathbf{X}) = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T$$

$$v = \text{Var}(Z) = \frac{1}{n} \sum_{i=1}^n (z_i - \gamma)^2$$

$$v = g(p, \mathbf{C})$$


Evaluating Variance of $\mathbf{Z}$

$$v = \frac{1}{n} \sum_{i=1}^n (z_i - \gamma)^2 = \frac{1}{n} \sum_{i=1}^n \underbrace{(z_i - \gamma)}_{\substack{\text{red arrow} \\ \text{to } p^T x_i - p^T \mu}} \underbrace{(z_i - \gamma)}_{\substack{\text{red arrow} \\ \text{to } x_i^T p - \mu^T p}}$$

$$v = \frac{1}{n} \sum_{i=1}^n (p^T x_i - p^T \mu)(x_i^T p - \mu^T p)$$

Evaluating Variance of **Z**

$$v = \frac{1}{n} \sum_{i=1}^n (\mathbf{p}^T \mathbf{x}_i - \mathbf{p}^T \boldsymbol{\mu})(\mathbf{x}_i^T \mathbf{p} - \boldsymbol{\mu}^T \mathbf{p})$$

$$v = \mathbf{p}^T \left\{ \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i^T - \boldsymbol{\mu}^T) \right\} \mathbf{p}$$

Pre-multiplier

Post-multiplier

Evaluating Variance of $\mathbf{Z}$

$$v = \mathbf{p}^T \left\{ \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i^T - \boldsymbol{\mu}^T) \right\} \mathbf{p}$$

$$v = \mathbf{p}^T \left\{ \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T \right\} \mathbf{p}$$

$$v = \mathbf{p}^T \mathbf{C} \mathbf{p}$$

Recapitulation: KKT Conditions

Minimize: $f(\mathbf{x})$ Subject to: $h_i(\mathbf{x}) = 0; i = 1, \dots, n$

Lagrangian: $g(\mathbf{x}; \alpha_1, \dots, \alpha_n) = f(\mathbf{x}) + \sum_{i=1}^n \alpha_i h_i(\mathbf{x})$

KKT Conditions: $\nabla_{\mathbf{x}} g(\mathbf{x}; \alpha_1, \dots, \alpha_n) = \nabla_{\mathbf{x}} f(\mathbf{x}) + \sum_{i=1}^n \alpha_i \nabla_{\mathbf{x}} h_i(\mathbf{x}) = \mathbf{0}$

Maximize: $f(\mathbf{x})$ Subject to: $h_i(\mathbf{x}) = 0; i = 1, \dots, n$

Lagrangian: $g(\mathbf{x}; \alpha_1, \dots, \alpha_n) = -f(\mathbf{x}) + \sum_{i=1}^n \alpha_i h_i(\mathbf{x})$

KKT Conditions: $\nabla_{\mathbf{x}} g(\mathbf{x}; \alpha_1, \dots, \alpha_n) = -\nabla_{\mathbf{x}} f(\mathbf{x}) + \sum_{i=1}^n \alpha_i \nabla_{\mathbf{x}} h_i(\mathbf{x}) = \mathbf{0}$

Problem Formulation: Step #1

Input $\mathbf{x}_i$ are projected along the direction $\mathbf{p}$. Maximize Variance of projected data $z_i = \mathbf{p}^T \mathbf{x}_i$ ($i = 1, \dots, n$). The optimal direction should be chosen while maintaining the prevailing constraint $\mathbf{p}^T \mathbf{p} = 1$.

$$g(\mathbf{p}; \alpha) = -\mathbf{p}^T \mathbf{C} \mathbf{p} + \alpha(\mathbf{p}^T \mathbf{p} - 1)$$

The Lagrangian

Solution: Step #1

The Lagrangian

$$g(\mathbf{p}; \alpha) = -\mathbf{p}^T \mathbf{C} \mathbf{p} + \alpha(\mathbf{p}^T \mathbf{p} - 1)$$

Using Karush-Kuhn-Tucker Conditions (KKT)

$$\nabla_{\mathbf{p}} g(\mathbf{p}; \alpha) = -2\mathbf{C} \mathbf{p} + \alpha(2\mathbf{p}) = \mathbf{0}$$



$$\mathbf{C} \mathbf{p} = \alpha \mathbf{p}$$

Step #1: Interpretation

$$\mathbf{C}\mathbf{p} = \alpha\mathbf{p}$$

The above Equation reminds us of the Relation between a Matrix and its Eigen Vector(s) and Corresponding Eigen Value(s). Here, the unit direction $\mathbf{p}$ is an Eigen Vector of $\mathbf{C}$ while α is its corresponding Eigen Value.

Step #1: Maximization

$$\begin{aligned} v &= \mathbf{p}^T \mathbf{C} \mathbf{p} = \mathbf{p}^T (\mathbf{C} \mathbf{p}) \\ &= \mathbf{p}^T (\alpha \mathbf{p}) = \alpha (\mathbf{p}^T \mathbf{p}) = \alpha \end{aligned}$$

The Maximum Variance is the Largest Eigen Value λ_{max} of the Covariance Matrix $\mathbf{C} = Cov(\mathbf{X})$ while the Optimal Direction of Projection is the corresponding Eigen Vector $\hat{\mathbf{e}}_{max}$ of $\mathbf{C}$

Problem Formulation: Step #2

Construction of Data Set $\mathbf{U}$ by Projection Along Direction $\mathbf{q}$

$$\mathbf{X} = \{\mathbf{x}_i : \mathbf{x}_i \in \mathbb{R}^D; i = 1, \dots, n\}$$

$$\mathbf{U} = \{u_i : u_i = \mathbf{q}^T \mathbf{x}_i; \forall \mathbf{x}_i \in \mathbf{X}; u_i \in \mathbb{R}^1\}$$

The Direction $\mathbf{q}$ is a Unit Vector

$$\mathbf{q}^T \mathbf{q} = 1$$

Data Sets $\mathbf{Z}$ and $\mathbf{U}$ are Uncorrelated

$$\mathbf{q}^T \hat{\mathbf{e}}_{max} = 0$$

Problem Formulation: Step #2

Input $\mathbf{x}_i$ are projected along the direction $\mathbf{q}$. Maximize Variance of projected data $u_i = \mathbf{q}^T \mathbf{x}_i$ ($i = 1, \dots, n$). The projected dataset $\mathbf{U}$ is uncorrelated with the other projected dataset $\mathbf{Z}$. The optimal direction $\mathbf{q}$ should be chosen while maintaining the following constraints.

$$(a) \mathbf{q}^T \mathbf{q} = 1 \text{ and } (b) \mathbf{q}^T \hat{\mathbf{e}}_{max} = 0$$

$$h(\mathbf{q}; \alpha, \beta) = -\mathbf{q}^T \mathbf{C} \mathbf{q} + \alpha(\mathbf{q}^T \mathbf{q} - 1) + \beta(\mathbf{q}^T \hat{\mathbf{e}}_{max})$$

The Lagrangian

Solution: Step #2

The Lagrangian

$$h(\mathbf{q}; \alpha, \beta) = -\mathbf{q}^T \mathbf{C} \mathbf{q} + \alpha(\mathbf{q}^T \mathbf{q} - 1) + \beta(\mathbf{q}^T \hat{\mathbf{e}}_{max})$$

Using Karush-Kuhn-Tucker Conditions (KKT)

$$\nabla_{\mathbf{q}} h(\mathbf{q}; \alpha, \beta) = -2\mathbf{C} \mathbf{q} + \alpha(2\mathbf{q}) + \beta \hat{\mathbf{e}}_{max} = 0$$



$$2\mathbf{C} \mathbf{q} = 2\alpha \mathbf{q} + \beta \hat{\mathbf{e}}_{max}$$

Solution: Step #2

$$2\mathbf{C}\mathbf{q} = 2\alpha\mathbf{q} + \beta\hat{\mathbf{e}}_{max}$$

Premultiply both sides by $\hat{\mathbf{e}}_{max}^T$

$$2\hat{\mathbf{e}}_{max}^T\mathbf{C}\mathbf{q} = 2\alpha(\underbrace{\hat{\mathbf{e}}_{max}^T\mathbf{q}}_{\hat{\mathbf{e}}_{max}^T\mathbf{q} = 0}) + \beta(\underbrace{\hat{\mathbf{e}}_{max}^T\hat{\mathbf{e}}_{max}}_{\hat{\mathbf{e}}_{max}^T\hat{\mathbf{e}}_{max} = 1})$$

$$\beta = 2\hat{\mathbf{e}}_{max}^T\mathbf{C}\mathbf{q}$$

Relation: $\mathbf{x}^T \mathbf{A} \mathbf{y} = \mathbf{y}^T \mathbf{A} \mathbf{x}$, If $\mathbf{A} = \mathbf{A}^T$

$$\mathbf{x} \in \mathbb{R}^D \quad \mathbf{A} \in \mathbb{R}^{D \times D} \quad \mathbf{y} \in \mathbb{R}^D \quad \rho \in \mathbb{R}^1$$

$$\rho = \mathbf{x}^T \mathbf{A} \mathbf{y} \Rightarrow \rho = (\mathbf{x}^T \mathbf{A} \mathbf{y})^T$$

$$\rho = \mathbf{x}^T \mathbf{A} \mathbf{y} = (\mathbf{x}^T \mathbf{A} \mathbf{y})^T = \mathbf{y}^T \mathbf{A}^T \mathbf{x} = \mathbf{y}^T \mathbf{A} \mathbf{x}$$

Solution: Step #2

$$\beta = 2\hat{\mathbf{e}}_{max}^T \mathbf{C} \mathbf{q}$$

$$\mathbf{x}^T \mathbf{A} \mathbf{y} = \mathbf{y}^T \mathbf{A} \mathbf{x}, \text{ if } \mathbf{A} = \mathbf{A}^T$$

$$\beta = 2\hat{\mathbf{e}}_{max}^T \mathbf{C} \mathbf{q} = 2\mathbf{q}^T \mathbf{C} \hat{\mathbf{e}}_{max} = 2\mathbf{q}^T (\mathbf{C} \hat{\mathbf{e}}_{max})$$

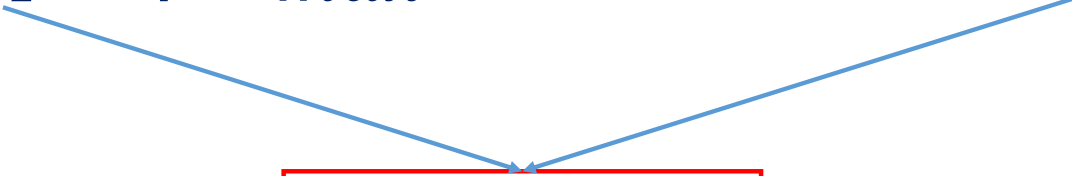
$$= 2\mathbf{q}^T (\lambda_{max} \hat{\mathbf{e}}_{max}) = 2\lambda_{max} (\mathbf{q}^T \hat{\mathbf{e}}_{max}) = 0$$

$$\boxed{\beta = 0}$$

Solution: Step #2

$$2\mathbf{C}\mathbf{q} = 2\alpha\mathbf{q} + \beta\hat{\mathbf{e}}_{max}$$

$$\beta = 0$$


$$\mathbf{C}\mathbf{q} = \alpha\mathbf{q}$$

The above Equation reminds us of the Relation between a Matrix and its Eigen Vector(s) and Corresponding Eigen Value(s). Here, the unit direction $\mathbf{q}$ is an Eigen Vector of $\mathbf{C}$ while α is its corresponding Eigen Value.

Step #2: Maximization

$$v = \mathbf{q}^T \mathbf{C} \mathbf{q} = \mathbf{q}^T (\mathbf{C} \mathbf{q}) = \mathbf{q}^T (\alpha \mathbf{q}) = \alpha (\mathbf{q}^T \mathbf{q}) = \alpha$$

For maximizing α , we can choose the Eigen Value next to λ_{max} . And, the Optimal Direction can be chosen as its corresponding Eigen Vector.

PCA: Dimensionality Reduction Transformation

Compute the Mean: $\boldsymbol{\mu} = \sum_{i=1}^n \mathbf{x}_i$

Zero-Centering of Input Data Set $\mathbf{X}$: $\mathbf{y}_i = \mathbf{x}_i - \boldsymbol{\mu}$

Compute the Covariance Matrix: $\mathbf{C} = \sum_{i=1}^n \mathbf{y}_i \mathbf{y}_i^T$

Eigen Values of $\mathbf{C}$ in Descending Order: $\boldsymbol{\Lambda} = [\lambda_1, \dots, \lambda_D]$

Corresponding Eigen Vectors of $\mathbf{C}$: $\mathbf{E} = [\hat{\mathbf{e}}_1, \dots, \hat{\mathbf{e}}_D]$

PCA: Constructing Transformation Matrix

Define $s_m = \sum_{i=1}^m \lambda_i$ ($\lambda_i \in \Lambda$)

Define $\rho_m = \frac{s_m}{s_D} = \frac{\lambda_1 + \dots + \lambda_m}{\lambda_1 + \dots + \lambda_m + \dots + \lambda_D}$

Choose *Eigen Vectors* Corresponding to *Top-k*
Eigen Values if $\rho_k \geq \eta$

$$\mathbf{E} = \underbrace{[\hat{\mathbf{e}}_1, \dots, \hat{\mathbf{e}}_k]}_Q | \hat{\mathbf{e}}_{k+1}, \dots, \hat{\mathbf{e}}_D]$$

PCA: Constructing Transformation Matrix

Transformation Matrix

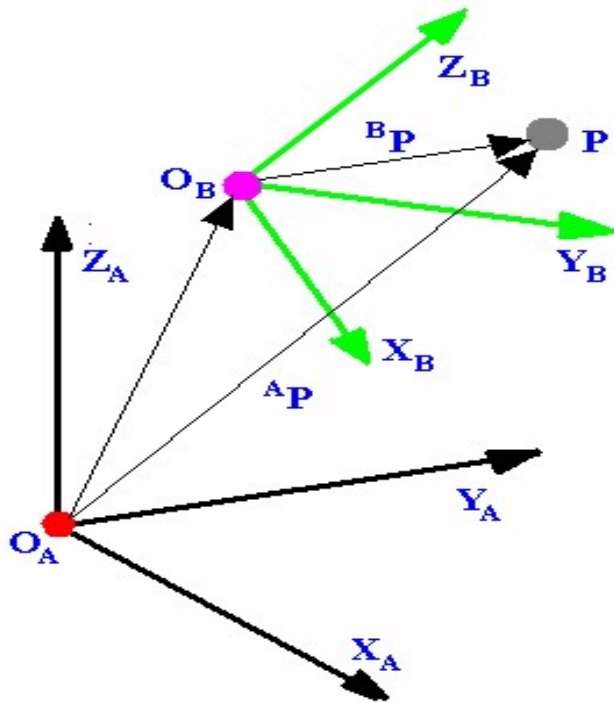
$$\mathbf{Q} = [\hat{\mathbf{e}}_1 \dots \hat{\mathbf{e}}_k]$$

Dimensionality Reduction Transformation

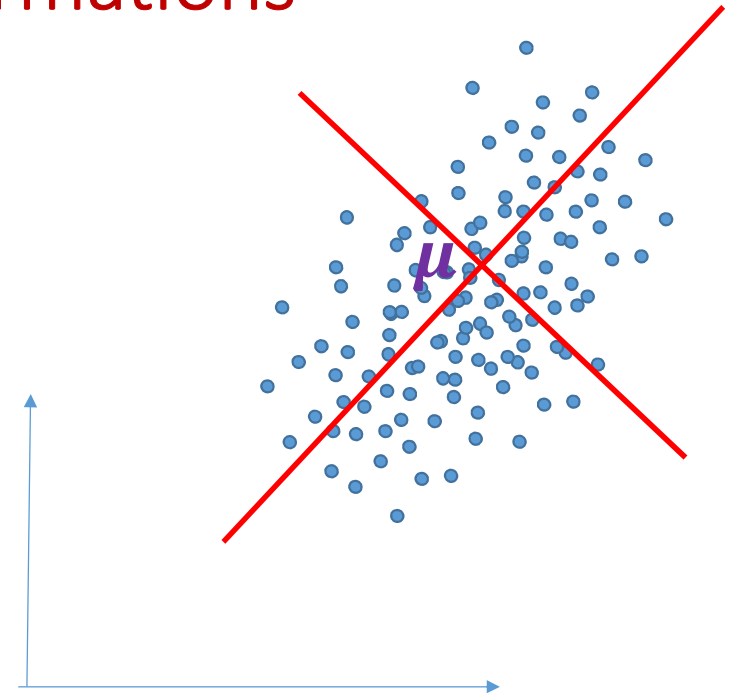
$$\mathbf{v} = \mathbf{Q}^T (\mathbf{x} - \boldsymbol{\mu})$$

$$\mathbf{v} \in \mathbb{R}^k \quad \mathbf{Q} \in \mathbb{R}^{D \times k} \quad \mathbf{x} \in \mathbb{R}^D \quad \boldsymbol{\mu} \in \mathbb{R}^D$$

Remembering Coordinate Transformations

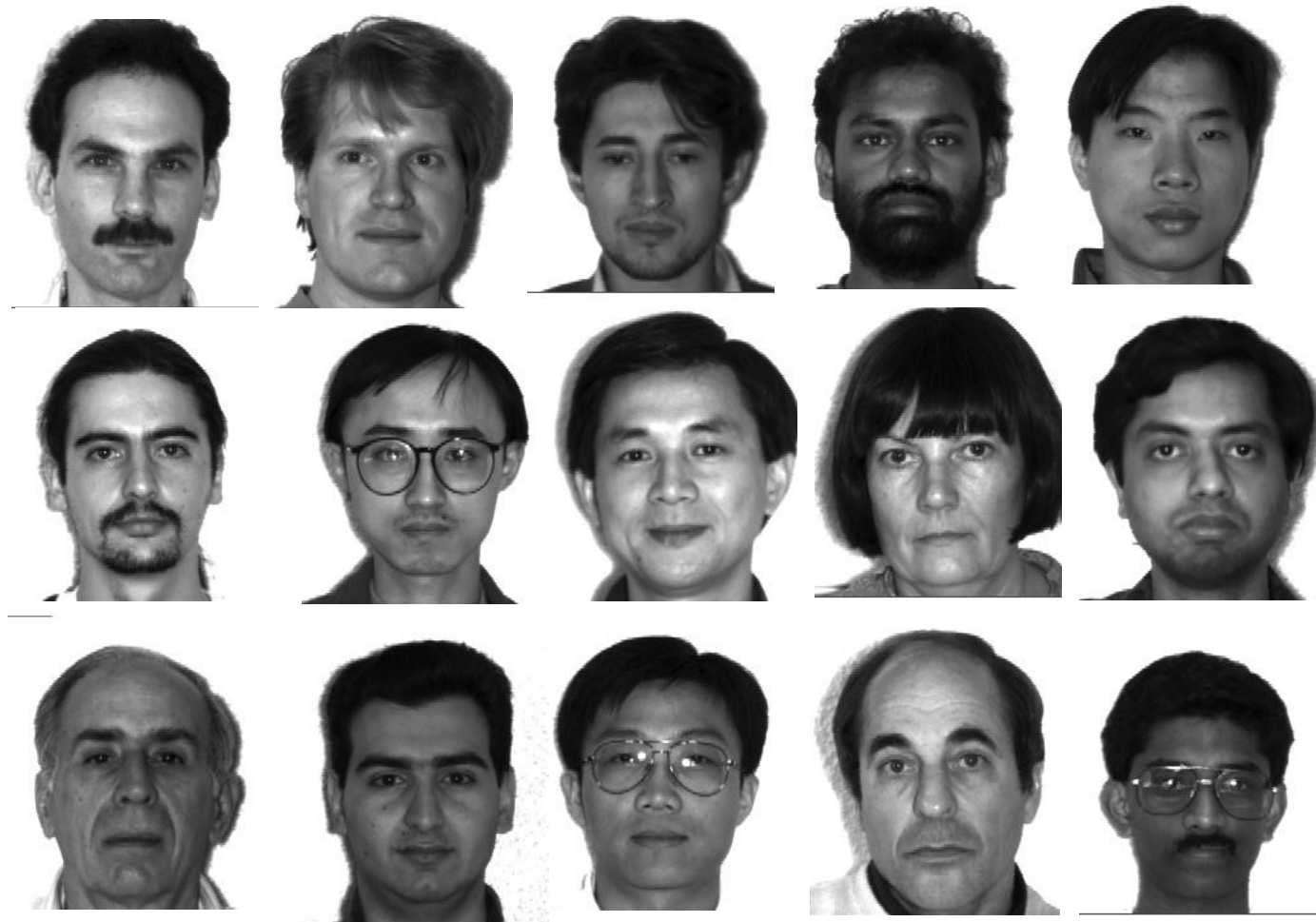


$${}^B P = {}_B^A R^T ({}^A P - {}^A O_B)$$

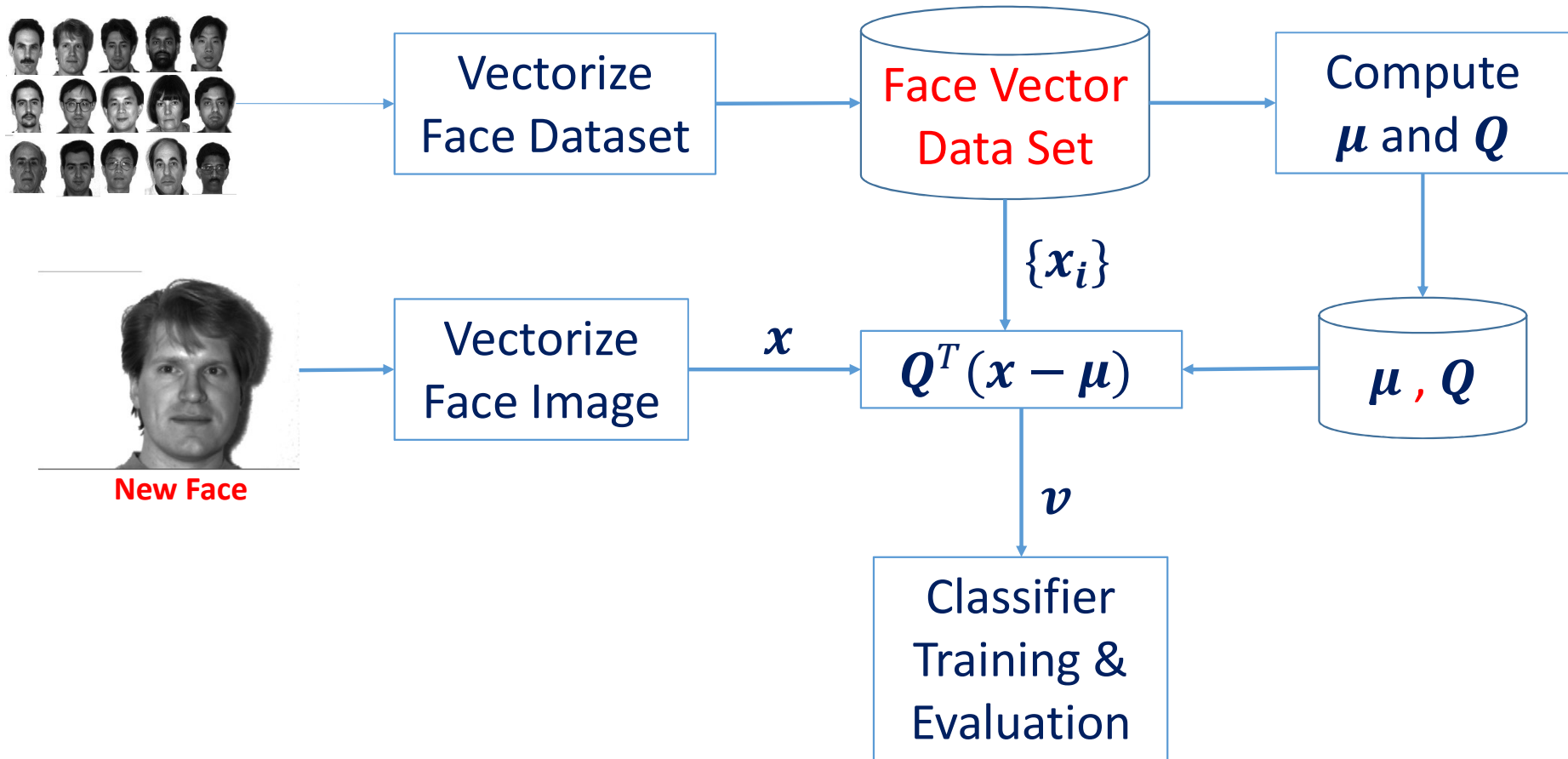


$$U = E^T (X - \mu)$$

Application: Face Feature Extraction



Summary



Summary

- Significance of Principal Components
- Eigen Vectors of Covariance Matrix
- Transformation Matrix from Eigen Vectors
- Provides Uncorrelated Dimensions
- Useful for Applications with
 - Highly Correlated Features
 - Features with Probably Redundant Information



Thank You